

Static and predictive tomographic reconstruction for wide-field multi-object adaptive optics systems

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Multi-object adaptive optics (MOAO) systems are still in their infancy: their complex optical designs for tomographic, wide-field wavefront sensing, coupled with open-loop (OL) correction, make their calibration a challenge. The correction of a discrete number of specific directions in the field allows for streamlined application of a general class of spatio-angular algorithms, initially proposed in Whiteley *et al.* [J. Opt. Soc. Am. A **15**, 2097 (1998)], which is compatible with partial on-line calibration. The recent Learn & Apply algorithm from Vidal *et al.* [J. Opt. Soc. Am. A **27**, A253 (2010)] can then be reinterpreted in a broader framework of tomographic algorithms and is shown to be a special case that exploits the particulars of OL and aperture-plane phase conjugation. An extension to embed a temporal prediction step to tackle sky-coverage limitations is discussed. The trade-off between lengthening the camera integration period, therefore increasing system lag error, and the resulting improvement in SNR can be shifted to higher guide-star magnitudes by introducing temporal prediction. The derivation of the optimal predictor and a comparison to suboptimal autoregressive models is provided using temporal structure functions. It is shown using end-to-end simulations of Raven, the MOAO science, and technology demonstrator for the 8 m Subaru telescope that prediction allows by itself the use of 1-magnitude-fainter guide stars. © 2013 Optical Society of America

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1. MULTI-OBJECT ADAPTIVE OPTICS

Single-object adaptive optics (AO) systems are now routinely used in most major observatories to attenuate the blurring effect of the Earth's atmosphere when imaging through turbulence. This blurring results in loss of angular resolution, set by the ratio of the imaging wavelength λ and the telescope diameter. Classical AO systems use a single wavefront sensor (WFS) and deformable mirror (DM) driven in real-time to measure and correct the wavefront phase aberrations in a single direction.

To achieve correction over a large field, angular anisoplanatism must be overcome. This phenomenon is linked to the variation of the optical disturbances across the field and is characterized by the anisoplanatic angle, $\theta_0 \propto \lambda^{6/5}$, typically of the order of tens of arcseconds in the H band [1].

Two approaches are envisioned for enlarging the isoplanatic angle: multi-object AO (MOAO) and multi-conjugate AO (MCAO). Both involve tomographic estimation of the 3D atmospheric wavefront disturbance using information from multiple WFSs locked on multiple guide stars, which probe different lines of sight through the atmosphere. The two approaches differ in how the correction is applied:

- In MCAO, one places multiple DMs in a series, each optically conjugated to a different atmospheric altitude. Due to the finite number of DMs, such systems will still suffer from generalized anisoplanatism [2].

- In MOAO, after the information from multiple WFSs is combined into a tomographic estimate of the turbulence [3], multiple science pick-off arms are placed on the scientifically interesting targets in the field. Each science channel contains a DM, which makes the optimal turbulence correction in its science direction. This parallel approach promises to increase the field over which AO corrections can be applied to 5 arcmin or even 10 arcmin.

However, unlike classical and MCAO closed loop AO systems, in which the WFSs sit after the DMs, MOAO systems require open-loop (OL) estimation of the atmosphere over a large field but only a few discrete number of correction directions. As such, the WFSs do not capture the DM shape, being bound to measure the total uncorrected disturbance (as opposed to measuring only the residual). The DM shape is not fed back to the system, being blindly applied to the device.

Because of its OL aspect, such systems stress the need for accurate calibration and call for (quasi) model-independent reconstructors. One solution for calibrating MOAO systems is the so-called Learn & Apply algorithm [4] in which the tomographic reconstructor is formulated as a minimum mean square error (MMSE) optimization problem, whose solution involves covariance matrices that can be directly measured in the MOAO system. These matrices encode a certain number of instrumental systematics and are, therefore, preferable to synthetic versions that may undermodel the actual phenomena in the real system.

A related MMSE solution had been previously proposed in Whiteley *et al.* [5] extending results of Valley [6] when dealing with optimal compensation of angular anisoplanatism in classical AO. As it turns out, MOAO can be seen as a generalization to multiple directions and any number of ground-conjugated correction directions. Therefore, a streamlined application of Whiteley's algorithm is formulated. Furthermore, the static Strehl-optimal reconstructors under consideration for 30–40 m class telescopes' closed-loop MCAO systems [7] can be worked out as variations (albeit mathematically equivalent) of the MMSE reconstructor.

As it stands, only partial computation of the tomographic reconstructor can be done from the data. The remainder, performing the angular extrapolation in the field from the guide stars to the science object direction(s), relies still on fitting an *a priori* atmospheric model to the data and cannot be circumvented.

Furthermore, to overcome temporal loop delays, or at least to mitigate them, the static reconstructors are then extended with an extra temporal prediction step. Potential predictive autoregressive (AR) models are compared to the one-step spatio-angular (SA) temporal solution under a common temporal structure-function analysis framework.

Temporal prediction has the potential to relax the temporal constraints, thus allowing for larger integration times and/or real-time processing. This may in turn increase the limiting magnitude of stars that can be used for guiding.

The performance analysis is done for Raven, the MOAO science and technology demonstrator for the 8 m Subaru telescope, supported by Monte Carlo simulations and experimental results from the Raven optical bench.

This paper has three main sections: in Section 2, two equivalent formulations are recalled and their features outlined. The explicit layered tomographic reconstructor allows more easily the formulation of an intermediate predictive step, whose options are discussed in Section 3. Finally, in Section 4, the gain in terms of limiting magnitude for the Raven demonstrator are presented and compared to numerical simulations.

2. SPATIAL, STREHL-OPTIMAL, STATIC RECONSTRUCTORS

Under the hypothesis that the turbulent atmosphere is a sum of N_l thin layers located in a discrete number of different altitudes h_j , the aperture-plane phase $\omega(\rho, \theta, t)$ indexed by the bi-dimensional spatial coordinate vector $\rho = (\rho_x, \rho_y)$ in direction $\theta = (\theta_x, \theta_y)$ at time t is defined as

$$\omega(\rho, \theta, t) = \sum_{j=1}^{N_l} \mathcal{W}_j(\rho + h_j \theta, t), \quad (1)$$

where $\mathcal{W}_j(\rho, t)$ is the j th-layer wave-front.

In the following, assume the phase is conveniently expanded onto a piston-removed Zernike orthonormal modal basis [8] with a finite number of N_z polynomials,

$$\omega(\rho, \theta, t) = \sum_{i=2}^{N_z} \phi_i(\theta, t) \mathbf{Z}_i(\rho/R), \quad (2)$$

with modal coefficients

$$\phi_i(\theta, t) = \frac{1}{R^2} \iint \mathbf{Z}_i(\rho/R) \omega(\rho, \theta, t) \Omega(\rho) d^2 \rho, \quad (3)$$

where $\Omega(\rho)$ is the aperture function, R the telescope primary mirror radius, and $\mathbf{Z}_i(\rho/R)$ the i th polynomial function.

Using matrix formulation, the resulting aperture-plane wavefront coefficients $\phi(\theta, t)$ in the near-field approximation relate to the coefficients φ of the wavefront phase defined over a discrete number of layers in the volume by a simple matrix multiplication [9]:

$$\phi(\theta, t) = \mathbf{P}_\theta \varphi(t), \quad (4)$$

where

$$\varphi \triangleq \begin{bmatrix} \varphi_0^T & \cdots & \varphi_{h_L}^T \end{bmatrix}^T \quad (5)$$

is a concatenation of phase coefficients of the decomposition of $\mathcal{W}_j(\rho, t)$, $1 \leq j \leq N_l$ onto the basis function set, and $\mathbf{P}_\theta$ is a cookie-cutter matrix that remaps and sums the Zernike coefficients from the wavefront modal expansion in the meta-pupils to the pupil-plane. The superscript T represents vector transpose.

As is common in AO, the aperture-plane phase is not measured directly. Instead, the WF is estimated (reconstructed) from some measured (and noisy) data that is statistically related to it. Consider only the widely used Hartmann–Shack wavefront sensor (HS-WFS), which provides the phase gradient $s = (s_x, s_y)$ with noise.

In MOAO, the objective cost function is the minimization of the aperture-plane residual phase variance (MV) for individual science directions $\beta_i \in \mathbb{R}^{1 \times 2}$:

$$\mathbf{E} = \arg \min_{\mathbf{E}} \langle \|\phi(\beta_i) - \hat{\phi}(\beta_i)\|_{L_2(\Omega)}^2 \rangle, \quad (6)$$

where ϕ is the actual phase (or its coefficients as is the case here), $\hat{\phi}(\beta_i) \triangleq \mathbf{E}s$ is the estimated phase (estimated quantities are indicated by the hat symbol), s are noisy measurements, L_2 is the Euclidean norm over the aperture Ω , and $\langle \cdots \rangle$ is the ensemble average over time for an individual optimization direction. As is shown in [10], the minimization is independent of the direction β_i , i.e., it is equivalent to minimize the residual phase variance over a given field-of-view (FoV) of interest. Due to the particular optimization for individual directions in MOAO, the reconstructors can be simplified, and a more compact solution is found as is described next.

A. Equivalent Tomographic Reconstructors

Let the following forward-measurement model, which makes explicit use of the layered phase vector [9,10],

$$s_\alpha(t) = \mathbf{\Gamma} \mathbf{P}_\alpha \varphi(t) + \eta(t) \quad (7)$$

with $s_\alpha(t) \in \mathbb{R}^{(N_\alpha \times N_s) \times 1}$ a column vector of $N_\alpha \times N_s$ measurements for all the N_α GS directions, obtained using the pupil-plane gradient operator $\mathbf{\Gamma} \triangleq \text{diag}\{\mathbf{\Gamma}_1, \dots, \mathbf{\Gamma}_{N_\alpha}\} \in \mathbb{R}^{N_\alpha N_s \times N_\alpha N_z}$ that concatenates N_α gradient operators for each individual WFS. $\mathbf{\Gamma}_i$ relates the N_z phase dimensions in the aperture in the directions $\alpha \in \mathbb{R}^{N_\alpha \times 2}$ to the N_s measurements. Noise is represented by $\eta(t) \triangleq \mathbb{R}^{N_\alpha N_s \times 1}$.

The MV reconstructor providing the aperture-plane phase estimate in the β science directions is obtained by minimizing the partial derivatives of Eq. (6) with respect to E' [7,10,11]:

$$\hat{\phi}_\beta = \mathbf{P}_\beta \mathbf{E} s_\alpha, \quad (8a)$$

$$= \mathbf{P}_\beta \langle \phi \phi^T \rangle \mathbf{P}_\alpha^T \Gamma^T (\Gamma \mathbf{P}_\alpha \langle \phi \phi^T \rangle \mathbf{P}_\alpha^T \Gamma^T + \langle \eta \eta^T \rangle)^{-1} s_\alpha, \quad (8b)$$

where $\phi_\beta \in \mathbb{R}^{N_\beta \times 1}$ is a vector of aperture-plane phase coefficients from the decomposition of the WF in all the N_β science directions $\beta \in \mathbb{R}^{N_\beta \times 2}$. This representation is preferred in closed loop wide-field AO systems and has been extensively used previously coupled to pseudo-OL control [12]. It is particularly suited to multi-conjugate AO, where the turbulence profile is needed prior to fitting to a DM's influence functions for correction. The deterministic fitting step is a least-squares fit to the DM influence functions that does not depend upon the measurement noise nor turbulence statistics.

With this formulation, the turbulence profile is explicitly estimated in the layers before being collapsed to the pupil plane through $\mathbf{P}_\beta$; it is therefore referred to as “explicit tomography reconstructor” in the remainder of this paper.

Before continuing, note that Eq. (8) is equivalent to the MMSE solution with a simplified measurement model involving the pupil-plane turbulence only:

$$s_\alpha(t) = \Gamma \phi_\alpha(t) + \eta(t). \quad (9)$$

Assuming s and ϕ are zero-mean and jointly Gaussian, direct application of the MMSE solution to estimate the aperture-plane phase in the N_β -science directions of interest yields [13]

$$\mathcal{E}\{\phi_\beta | s_\alpha\} \triangleq \Sigma_{(\phi_\beta, s_\alpha)} \Sigma_{s_\alpha}^{-1} s_\alpha = \hat{\phi}_\beta, \quad (10)$$

where $\mathcal{E}\{X|Y\}$ stands for the mathematical expectation of X conditioned to Y . Since in general $\beta \neq \alpha$, Eq. (10) follows from $\mathcal{E}\{\phi_\beta | s_\alpha\} = \mathcal{E}\{\phi_\beta | \mathcal{E}\{\phi_\alpha | s_\alpha\}\}$. Given that the conditioning relates only to the last available measurement (as opposed to present and previous measurements), these reconstructors are labeled as *static*. Developing terms in Eq. (10) using Eq. (9), the reconstructor becomes

$$\hat{\phi}_\beta \triangleq \langle \phi_\beta \phi_\alpha^T \rangle \Gamma^T (\Gamma \langle \phi_\alpha \phi_\alpha^T \rangle \Gamma^T + \langle \eta \eta^T \rangle)^{-1} s_\alpha. \quad (11)$$

Equation (11) converts to Eq. (8) by setting

$$\langle \phi_\beta \phi_\alpha^T \rangle = \mathbf{P}_\beta \langle \phi \phi^T \rangle \mathbf{P}_\alpha^T, \quad (12a)$$

$$\langle \phi_\alpha \phi_\alpha^T \rangle = \mathbf{P}_\alpha \langle \phi \phi^T \rangle \mathbf{P}_\alpha^T. \quad (12b)$$

This MMSE reconstructor is dubbed SA reconstructor, a term coined by Rodolphe Conan, to the best of the authors' knowledge, on account of the nature of the covariance matrices involved in its definition. It can be seen as a generalization of the work of Whiteley *et al.* [5] in seeking the optimal anisoplanatic reconstructor in classical AO to the tomographic, multiple sensor case. It has several convenient features: it is much faster to compute off-line than the explicit tomography reconstructor and circumvents the truncated expansion on a modal basis.

The numerical equivalence of Eq. (12) is further explored in Section 4.A for the Raven demonstrator [1].

When phase is expanded onto the Zernike polynomials, the SA cross-covariance functions can be analytically computed for the infinite outer-scale case of turbulence [14,15]:

$$\langle \phi_i(0) \phi_j(\xi) \rangle = 3.895 \left(\frac{D}{r_0} \right)^{\frac{5}{3}} \frac{\int_0^{h_{\max}} C_n^2(h) I_{ij} \left(\frac{\xi h}{R} \right) dh}{\int_0^\infty C_n^2(h) dh} \quad (13)$$

with $D = 2R$, the telescope diameter; r_0 , the Fried parameter; h , the altitude; ξ , the angle between the pupils over which the Zernike polynomials are defined; C_n^2 , the atmospheric vertical profile; and $I_{ij}(x)$, a term involving 1D numerical integration. Equation (13) has been extended for the finite outer-scale case in [16] and later extensively used and generalized in [5,17]. The layered spatial covariance matrix $\langle \phi \phi^T \rangle$ from Eq. (8b) is a block-diagonal matrix (layers are independent) and can be found in [8] for the infinite outer-scale case and in [18] for the finite case.

B. Spatio Angular versus Explicit Tomography

OL operation of MOAO systems poses a complex problem for system calibration. MOAO is exposed to potential issues, such as misregistration, field-dependent distortions, and irregular sensitivity (to cite a few) since the WFSs do not “see” the actual DM figures. In closed-loop systems, the recursive nature of the loop using feedback allows for partial compensation of miscalibrations.

The SA formulation is thus particularly amenable to MOAO systems since, at least in principle, the covariance matrices composing the reconstructor can be directly computed from acquired data. This could not be achieved in a closed-loop system. In regular on-sky operation, Σ_{s_α} from Eq. (10) can be recorded on-line (with use of some caution to ensure statistical convergence and proper acquisition). This covariance matrix provides a wealth of valuable information about the system itself with the strong potential to carry the signature of misalignments and other spatially variant discrepancies not taken into account in the models. Also, and equally important, it carries an imprint of the vertical turbulence profile, from which the C_n^2 profile and integrated atmospheric parameters estimation, such as r_0 , L_0 , and eventually the wind velocity vectors, can be estimated through data postprocessing using the built-in SLODAR method [19]. These parameters are then used to constrain a model for the second covariance matrix.

The Learn & Apply algorithm from [4] can now be presented as the SA reconstructor with covariances defined in slope space instead of phase space. The choice for remaining in slope space is due to system calibration; the algorithm directly collects measurements' covariances and makes the angular extrapolation in measurement space, bypassing an explicit slopes-to-phase reconstruction. Thus the projection onto and from phase space is circumvented at the expense of having larger covariance matrices (there are roughly twice as many slopes as estimated phase vectors).

Provided access to the layered turbulence is granted, linear prediction could be plugged into the explicit tomography reconstructor as an intermediary step to counter intrinsic temporal delays in the AO system, from integration and processing. The OL working environment excludes dynamical stability issues associated to feedback systems. Using Taylor's

frozen-flow approximation, the wavefront on each layer can be predicted before collapsing it onto the pupil-plane in the directions of interest. Options are more broadly investigated in Section 3.

3. PREDICTIVE MODELS FOR ATMOSPHERIC RECONSTRUCTION

As is well known for any AO system, the availability of sufficiently bright guide-stars overshadows the utility of AO. Therefore the sky coverage, i.e., the percentage of available sky for observing, imposes a strong constraint on the number of observable targets. Sky coverage can be improved by either making better use of every single photon (better wavefront sensing) or by relaxing the temporal lag-error constraint, which allows for longer integration times if the error can be partly overcome by temporal prediction. Lag error is intrinsic to any AO system due to the discrete nature of the measurements, their processing and correction. All combined requires several milliseconds to read-out and compute which stresses the importance of a predictive capability. The lag error definition adopted here is the sum of the integration time T_s plus the pure-delay Δ error encompassing the real-time processing of measurements and computation of DM commands. See the temporal diagram in Fig. 1.

When Δ is not an integer multiple of the frame-rate measurements, commands become asynchronous. Working in OL greatly simplifies the implementation when compared to closed-loop systems since in the measurement model the asynchronous DM commands do not intervene. The latter are computed as the weighted average over two consecutive phase instances spanned by the DM commands, using the fractional delays as weights. Such procedure would considerably increase the computational complexity, in particular when prediction is on. For this reason, in the remainder the case, $\Delta \neq 0$ is considered in the simulations but is not taken into account on the reconstructors.

A. Beating Down Temporal Lag Errors

As previously stated, the explicit-tomography formulation is well suited to temporal prediction. Note, though, that with some extra complexity, the same could be built into the SA formulation [5], albeit with no flexibility as to the choice of the predictive model.

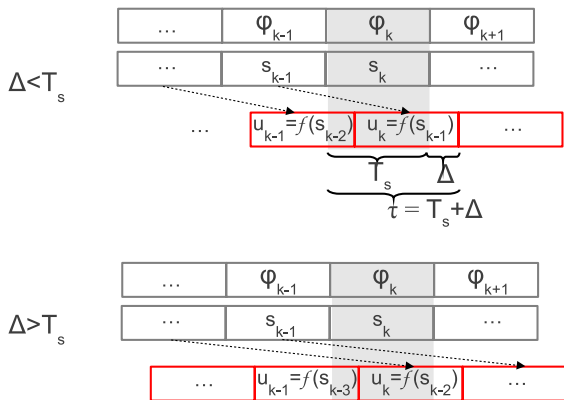


Fig. 1. Temporal diagrams. Top: $\text{Int}(\Delta/T_s) = 0$ commands u_k are conditioned to measurement s_{k-1} . Bottom: $T_s < \Delta < 2T_s$; u_k is conditioned to s_{k-2} .

Proposition

Temporal prediction can be achieved by means of a linear operation $\mathcal{P}$ on current and past samples of the phase vectors, such that

$$\hat{\phi}_{\beta_j, k+1} = \mathbf{P}_{\beta_j} \mathcal{P}[\hat{\phi}_k, \hat{\phi}_{k-1}, \dots, \hat{\phi}_{k-n}], \quad (14)$$

where $\mathcal{P}$ is a linear prediction operator. For the case of no prediction, $\mathcal{P} = \mathbf{I}$.

In the following, Taylor's frozen-flow hypothesis is assumed:

$$\omega(\rho, t + \tau) = \omega(\rho + \mathbf{v} \cdot \tau, t). \quad (15)$$

A first approach uses the fact that the spatial gradient of phase is measured by the HS-WFS. As a consequence, with a truncated polynomial expansion of Eq. (15) to the first-order [20], turbulence could be predicted. However, this model is restricted to single-conjugated AO systems and lacks generality for wide-field tomographic phase estimation—unless layer-oriented tomography is used [21], a framework not adopted here.

B. One-Step SA Predictor

A more general method that complies with tomographic phase estimation is sought. Time evolution of expansion coefficients leads to non-null cross correlations [17], which should be fully grasped to obtain an optimal prediction. Figure 2 depicts the modal cross-correlation functions for tip and focus (polynomials 2 and 4, respectively, following [8]) up to the 9th mode recurring to Eq. (13).

Recasting the problem as a proper criterion minimization, the best linear predictor is the solution to the following criterion:

$$\mathcal{A}_\tau = \arg \min_{\mathcal{A}_\tau} \langle \|\phi(t + \tau) - \mathcal{A}_\tau \phi(t)\|_{L_2(\Omega)}^2 \rangle, \quad (16)$$

yielding

$$\mathcal{A}_\tau \triangleq \langle \phi_{k+1} \phi_k^T \rangle \langle \phi_k \phi_k^T \rangle^{-1}. \quad (17)$$

This operation in what follows is called a “one-step SA predictor” (the face-on pattern of $\mathcal{A}_\tau$ is depicted in Fig. 3). It is a nondiagonal, densely populated matrix; a sign of the temporal cross-correlations between modes coming into play. The same predictor is outlined by [11] for the case of phase represented by its samples on a regular grid of points (also called the zonal representation).

The one-step cross-covariance matrix $\langle \phi_{k+1} \phi_k^T \rangle$ can be easily estimated from Eq. (13), under the assumption of isotropic and stationary turbulence. However, it has weak dependence on the wind direction (see, for instance, the short-term auto-correlation for modes tip and tilt in Fig. 4). Therefore, for pure frozen flow, a modal approach can be short of a high-fidelity phase-shifting approach, although this could be done at the expense of extra computation.

Furthermore, Eq. (17) is a general method to generate and predict phase in a 2D plane (any wind velocity can be used), according to the Markovian model [13]:

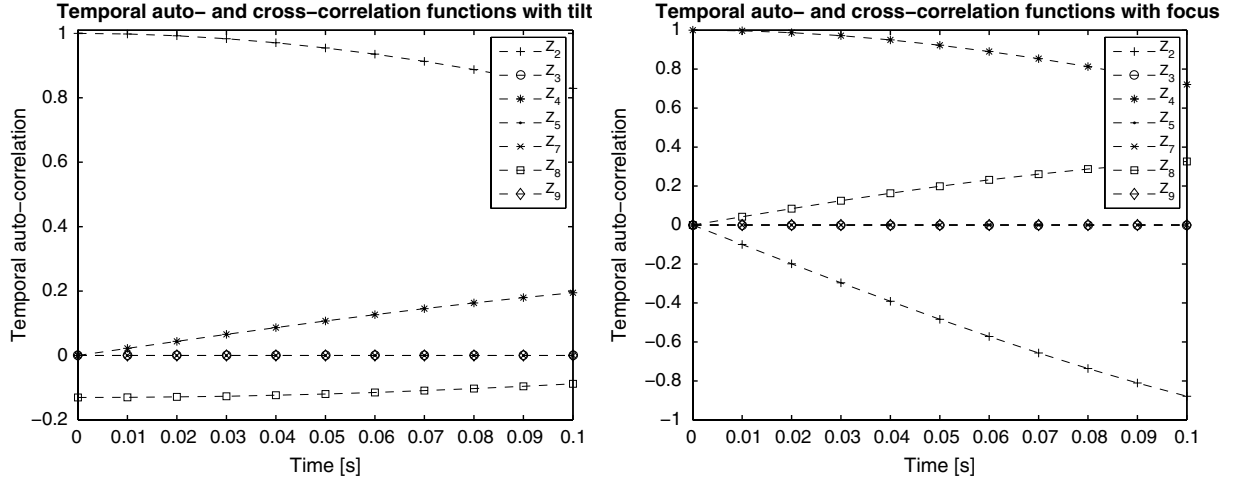


Fig. 2. Temporal auto- and cross-correlation functions for the tip and focus modes. Although at $t = 0$, these plots show the spatial values in [8], for $t > 0$ correlations appear and vanish as shown. One can see, for the case of focus, that although it isn't correlated to any other mode plotted for $t = 0$, a strong (anti-) correlation appears with tilt as time elapses.

$$\boldsymbol{\varphi}_{k+1} = \mathcal{A}_\tau \boldsymbol{\varphi}_k + \boldsymbol{\varepsilon}_k^\delta, \quad (18)$$

where $\boldsymbol{\varepsilon}_k^\delta$ is an excitation noise whose properties are fixed to guarantee proper turbulence statistics.

This model can either be used for generating a fully developed turbulence or for control-oriented purposes such as a Kalman filter gain computation [13]. In either case, the pair $(\mathcal{A}_\tau, \Sigma_\varepsilon^\delta)$ is required. The excitation noise covariance matrix $\Sigma_\varepsilon^\delta$ is found from imposing the output statistics to be those of a Kolmogorov or von-Kármán model. Hence $\Sigma_\varepsilon^\delta = \Sigma_\varphi - \mathcal{A}_\tau \Sigma_\varphi \mathcal{A}_\tau^T$ since $\langle \boldsymbol{\varphi}_{k+1} \boldsymbol{\varphi}_{k+1}^T \rangle = \mathcal{A}_\tau \Sigma_\varphi \mathcal{A}_\tau^T + \Sigma_\varepsilon^\delta$. Note $\Sigma_\varphi \triangleq \langle \boldsymbol{\varphi}_{k+1} \boldsymbol{\varphi}_{k+1}^T \rangle \triangleq \langle \boldsymbol{\varphi}_k \boldsymbol{\varphi}_k^T \rangle$, which, due to stationarity, loses its temporal dependence.

A further remark is that with such Markovian processes (future values are only conditioned to present ones), a simplification can be done. It consists of considering the total delay $\tau = T_s + \Delta m$ simplicity as the sampling step of the predictive model. The asynchronous case is thus dealt with efficiently with no recourse to weighted averaging.

Assémat's method for simulating infinitely long, nonstationary phase screens [22] turns out to be a truncated version of

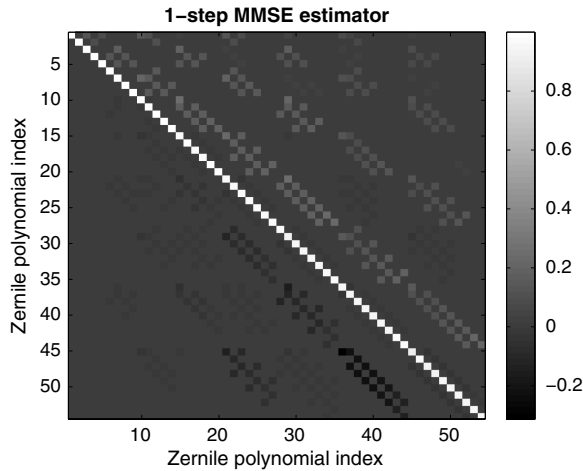


Fig. 3. Face-on pattern of $\mathcal{A}_\tau$, single-layer, computed for nine radial orders ($N_z = 55$) with $|v| = 15 \text{ ms}^{-1}$ and $\tau = 10 \text{ ms}$, from which results a displacement of 0.15 m .

Eq. (17) when a pointwise (zonal) representation of the phase is used. However, only a subset of the columns of $\mathcal{A}_\tau$ is used to include bounded region correlations. Extensions can be found in [23,24].

Despite the temporal cross-correlation being taken into account in this model, $\mathcal{A}_\tau$ is strongly diagonally dominated, suggesting that simpler diagonal models, i.e., mode-by-mode, could be potentially applied, and indeed they have been extensively used in AO simulations in the form of AR models [25]. Their features and relation to the SA predictor are explored next.

C. Autoregressive Models

For several AO applications constrained by real-time computational burden, using simpler diagonal AR models can be quite appealing as they circumvent $\mathcal{A}_\tau$, being a dense matrix. Although these relatively coarse models are not adapted to simulating atmospheric turbulence, they are used instead for prediction when embedded in the reconstructor and plugged into the controller—as is done with Kalman filtering [26,27]—for off-line computation of optimal gains.

An AR model of order n is defined by the recursion

$$\boldsymbol{\varphi}_{k+1} = f(\boldsymbol{\varphi}_k, \dots, \boldsymbol{\varphi}_{k-n+1}) + \boldsymbol{\varepsilon}_k, \quad (19)$$

where $f(\dots)$ is a linear function yet to be defined, and $\boldsymbol{\varepsilon}_k$ is a Gaussian-distributed spectrally white zero-mean random sequence with variance such that the output variance is conserved as in the previous section.

An AR model of the first order (AR1) is simply

$$\boldsymbol{\varphi}_{k+1}^{\text{AR1}} = \mathbf{A}_{\text{AR1}} \boldsymbol{\varphi}_k^{\text{AR1}} + \boldsymbol{\varepsilon}_k^{\text{AR1}}, \quad (20)$$

where a diagonal $\mathbf{A}_{\text{AR1}}$ replaces $\mathcal{A}_\tau$ in Eq. (18). The excitation noise covariance matrix for the model in Eq. (20) is likewise computed as in Section 3.B.

For a second-order model (AR2):

$$\boldsymbol{\varphi}_{k+1}^{\text{AR2}} = \mathbf{A}_{\text{AR2}} \boldsymbol{\varphi}_k^{\text{AR2}} + \mathbf{B}_{\text{AR2}} \boldsymbol{\varphi}_{k-1}^{\text{AR2}} + \boldsymbol{\varepsilon}_k^{\text{AR2}}. \quad (21)$$

Imposing $\Sigma_\varphi^{\text{AR2}} = \Sigma_\varphi$ as before, $\Sigma_\varepsilon^{\text{AR2}}$ is found from

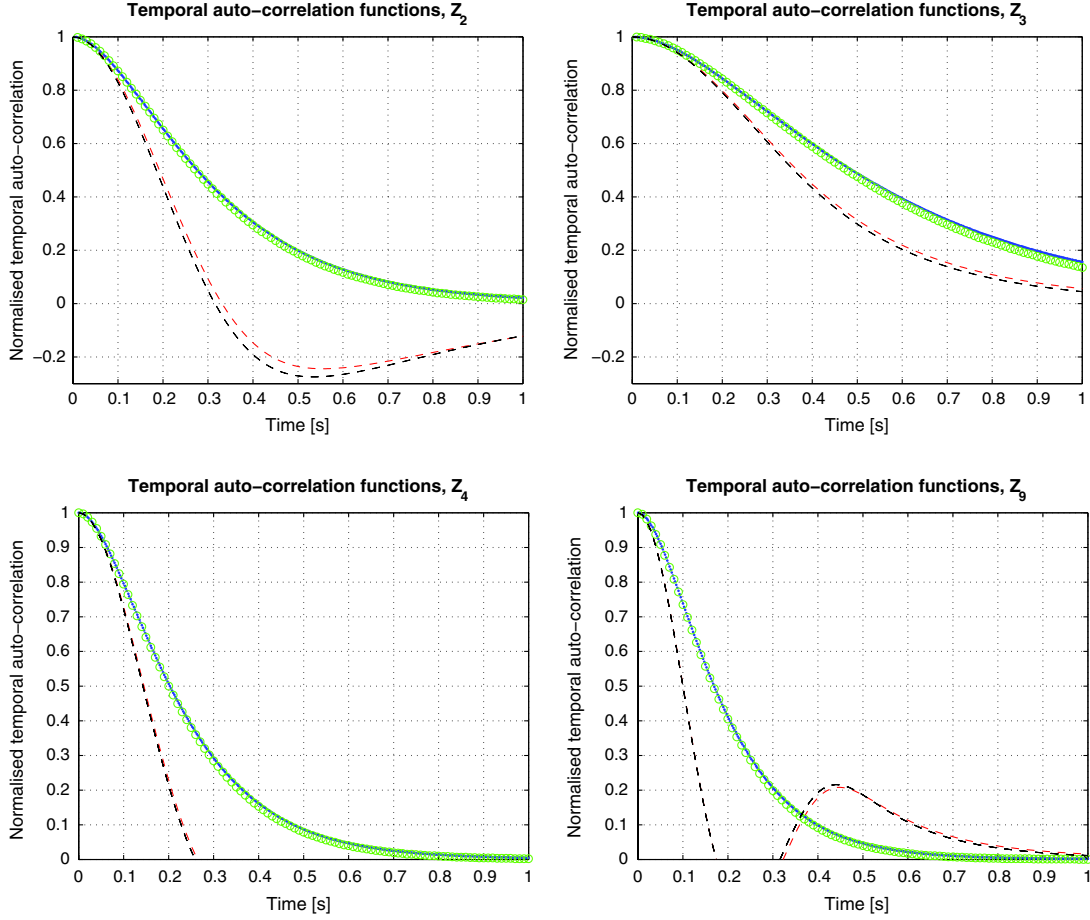


Fig. 4. Theoretical temporal auto-correlation functions assuming frozen flow computed directly (black dash) or as the Fourier-transformed temporal PSDs (red dash). Comparison against the second-order continuous (blue dots) and discrete (green circles) predictive models fitting the initial 50 ms of the theoretical curves. Model fitting uses the Broyden–Fletcher–Goldfarb–Shanno method.

$$\begin{aligned} \Sigma_{\varepsilon}^{\text{AR2}} = & \Sigma_{\varphi} - \mathbf{A}_{\text{AR2}} \Sigma_{\varphi} \mathbf{A}_{\text{AR2}}^T - \mathbf{B}_{\text{AR2}} \Sigma_{\varphi} \mathbf{B}_{\text{AR2}}^T - \mathbf{A}_{\text{AR2}} \Sigma_{1\tau}^{\text{AR2}} \mathbf{B}_{\text{AR2}}^T \\ & - \mathbf{B}_{\text{AR2}} \Sigma_{1\tau}^{\text{AR2}} \mathbf{A}_{\text{AR2}}^T, \end{aligned} \quad (22a)$$

$$\Sigma_{1\tau}^{\text{AR2}} = \langle \varphi_{k+1}^{\text{AR2}} (\varphi_k^{\text{AR2}})^T \rangle \triangleq \langle \varphi_k^{\text{AR2}} (\varphi_{k-1}^{\text{AR2}})^T \rangle = (\mathbf{I} - \mathbf{B}_{\text{AR2}})^{-1} \mathbf{A}_{\text{AR2}} \Sigma_{\varphi}, \quad (22b)$$

where $\Sigma_{1\tau}^{\text{AR2}}$ is a one-step modal spatiotemporal cross-covariance matrix.

Similarly, for a third-order model (AR3):

$$\varphi_{k+1}^{\text{AR3}} = \mathbf{A}_{\text{AR3}} \varphi_k^{\text{AR3}} + \mathbf{B}_{\text{AR3}} \varphi_{k-1}^{\text{AR3}} + \mathbf{C}_{\text{AR3}} \varphi_{k-2}^{\text{AR3}} + \varepsilon_k^{\text{AR3}}, \quad (23)$$

the noise covariance matrix is found from

$$\begin{aligned} \Sigma_{\varepsilon}^{\text{AR3}} = & \Sigma_{\varphi} - \mathbf{A}_{\text{AR3}} \Sigma_{\varphi} \mathbf{A}_{\text{AR3}}^T - \mathbf{B}_{\text{AR3}} \Sigma_{\varphi} \mathbf{B}_{\text{AR3}}^T - \mathbf{C}_{\text{AR3}} \Sigma_{\varphi} \mathbf{C}_{\text{AR3}}^T \\ & - \mathbf{A}_{\text{AR3}} \Sigma_{1\tau}^{\text{AR3}} \mathbf{B}_{\text{AR3}}^T - \mathbf{B}_{\text{AR3}} \Sigma_{1\tau}^{\text{AR3}} \mathbf{A}_{\text{AR3}}^T - \mathbf{B}_{\text{AR3}} \Sigma_{1\tau}^{\text{AR3}} \mathbf{C}_{\text{AR3}}^T \\ & - \mathbf{C}_{\text{AR3}} \Sigma_{1\tau}^{\text{AR3}} \mathbf{B}_{\text{AR3}}^T - \mathbf{A}_{\text{AR3}} \Sigma_{2\tau}^{\text{AR3}} \mathbf{C}_{\text{AR3}}^T - \mathbf{C}_{\text{AR3}} \Sigma_{2\tau}^{\text{AR3}} \mathbf{A}_{\text{AR3}}^T, \end{aligned} \quad (24a)$$

where $\Sigma_{1\tau}^{\text{AR3}}$ is again the one-step modal cross-covariance matrix; this time for the AR3 model, and $\Sigma_{2\tau}^{\text{AR3}} \triangleq \langle \varphi_{k+1}^{\text{AR3}}$

$(\varphi_{k-1}^{\text{AR3}})^T \rangle \triangleq \langle \varphi_k^{\text{AR3}} (\varphi_{k-2}^{\text{AR3}})^T \rangle$ is a two-step modal spatiotemporal cross-covariance matrix, with

$$\Sigma_{1\tau}^{\text{AR3}} = (\mathbf{B}_{\text{AR3}} + \mathbf{C}_{\text{AR3}} \mathbf{A}_{\text{AR3}} + \mathbf{C}_{\text{AR3}}^2)^{-1} (\mathbf{A}_{\text{AR3}} + \mathbf{C}_{\text{AR3}} \mathbf{B}_{\text{AR3}}) \Sigma_{\varphi}, \quad (24b)$$

$$\Sigma_{2\tau}^{\text{AR3}} = \mathbf{B}_{\text{AR3}} \Sigma_{\varphi} + (\mathbf{A}_{\text{AR3}} + \mathbf{C}_{\text{AR3}} \mathbf{B}_{\text{AR3}}) \Sigma_{1\tau}^{\text{AR3}}. \quad (24c)$$

1. Model Identification Using Zernike Polynomials

A practical method to identify matrices $\mathbf{A}$, $\mathbf{B}$, and $\mathbf{C}$ is to fit the initial T_{fit} seconds of the temporal auto-correlation function of each and every mode as previously suggested in [28]. An alternative is to match the decorrelation at the coherence time of each mode [29]. For AR1 models, both strategies lead to roughly the same modal decorrelation functions and can therefore be used interchangeably. For higher-order models (AR2 and AR3), fitting the initial T_f seconds leads to the overall best prediction performances, although the existence of a general T_{fit} is debatable and subject to optimization.

The temporal auto-correlation function is the Fourier-transformed temporal power spectral density (PSD), a direct application of the Wiener–Khinchine theorem (on a mode-by-mode basis). The PSDs are computed from [30].

A much faster and effective method builds on the SA covariance matrices of Eq. (13), with an angle given by $\xi = \tau \times |v|/h$ (under Taylor's frozen-flow hypothesis). This method is much more cost-effective since the correlation functions are computed directly and only over the T_{fit} horizon and not beyond, leading to considerable computational gains.

Figure 4 plots the temporal auto-correlation functions from Zernike polynomials 1 to 9 using the Wiener–Kinchine theorem against the SA formulation. The numerical agreement is remarkable, with slight discrepancies due to the limits of integration in either case. Also over-plotted are the second-order continuous model and its AR discretization for $T_s = 10$ ms, when fitting the initial $T_{\text{fit}} = 50$ ms. The AR (discrete time) model is obtained using standard zero-order-hold discretization procedures.

D. Prediction Error Structure Functions

The predictive capabilities of different models can be assessed using the predicted phase temporal structure function. Define the temporal lag error (no prediction) as the average phase variance taken two instants apart, i.e., the temporal structure function of the phase,

$$D_t(\tau) = \langle |\omega(\rho, t) - \omega(\rho, t + \tau)|^2 \rangle, \quad (25)$$

which relates to the spatial structure function of phase D_ρ through Taylor's frozen-flow hypothesis by

$$D_t(\tau) = \langle |\omega(\rho, t) - \omega(\rho - v \cdot \tau, t)|^2 \rangle_\rho = D_\rho(v \cdot \tau), \quad (26)$$

with $\langle \dots \rangle_\rho$ the ensemble average over the spatial distances $\rho = |\rho|$, $v = |v|$. Expanding the squared term and assuming stationarity gives $2(C(0) - C_t(\tau)) = 2(C(0) - C_\rho(v\tau))$, where $C_t(\cdot)$ and $C_\rho(\cdot)$ are the phase temporal and spatial covariance functions. $C_t(\tau) = C_\rho(v\tau)$, i.e., the temporal covariance function is deduced from the phase spatial covariance function evaluated at $\rho = v\tau$.

The phase spatial structure function is given by [6]

$$D_\rho(\rho) = \left(\frac{L_0}{r_0}\right)^{5/3} \times \frac{2^{1/6}\Gamma(11/6)}{\pi^{8/3}} \left[\frac{24}{5} \Gamma\left(\frac{6}{5}\right) \right]^{5/6} \times \left[\frac{\Gamma(5/6)}{2^{1/6}} - \left(\frac{2\pi\rho}{L_0}\right)^{5/6} K_{5/6}\left(\frac{2\pi\rho}{L_0}\right) \right], \quad (27)$$

with L_0 the outer scale of turbulence, r_0 Fried's coherence length, Γ the "gamma" function, and $K_{5/6}$ a modified Bessel function of the third kind.

It is useful to consider the general formulation for the temporal lag error that is compatible with the case of atmospheric prediction:

$$\sigma_{\text{lag}}^2(\tau, p) = \langle \|\mathbf{P}_\theta(\boldsymbol{\varphi}_k - \hat{\boldsymbol{\varphi}}_k)\|_{L_2(\Omega)}^2 \rangle, \quad (28)$$

where $\hat{\boldsymbol{\varphi}}_k$ is the phase estimate using any of the predictive models of order $p = \{0, 1, 2, 3\}$.

In the **no** prediction case, the estimated phase is simply a replication of the phase at the previous time step, $\hat{\boldsymbol{\varphi}}_k = \boldsymbol{\varphi}_{k-1}$. The temporal lag error from Eq. (28) becomes

$$\sigma_{\text{lag}}^2(\tau, p = 0) = 2 \text{trace}\{\mathbf{P}_\theta(\Sigma_\varphi - \Sigma_{1\tau})\mathbf{P}_\theta^T\} \triangleq D_t(\tau), \quad (29)$$

which is the temporal structure function of phase from Eq. (26), with the one-step covariance matrix $\Sigma_{1\tau} = \Sigma(\rho = v\tau)$ computed from Eq. (13) with a proper angle.

These temporal structure functions can now be expanded for the case of predicted phase. For first-order models (the AR1 or the one-step SA predictor) one has $\hat{\boldsymbol{\varphi}}_k = \mathbf{A}\boldsymbol{\varphi}_{k-1}$ (note the excitation noise is not included). Developing Eq. (28) yields

$$\sigma_{\text{lag}}^2(\tau, p = 1) = \text{trace}\{\mathbf{P}_\theta(\Sigma_\varphi + \mathbf{A}\Sigma_\varphi\mathbf{A}^T - 2\Sigma_{1\tau}\mathbf{A}^T)\mathbf{P}_\theta^T\}, \quad (30)$$

with $\mathbf{A} = \mathbf{A}_{\text{AR1}}$ for the AR1 model and $\mathbf{A} = \mathcal{A}_\tau$ for the SA predictor model. For the AR2:

$$\sigma_{\text{lag}}^2(\tau, p = 2) = \text{trace}\{\mathbf{P}_\theta(\Sigma_\varphi + \mathbf{A}_{\text{AR2}}\Sigma_\varphi\mathbf{A}_{\text{AR2}} + \mathbf{B}_{\text{AR2}}\Sigma_\varphi\mathbf{B}_{\text{AR2}} - 2\mathbf{A}_{\text{AR2}}\Sigma_{1\tau} + \mathbf{A}_{\text{AR2}}\Sigma_{1\tau}\mathbf{B}_{\text{AR2}}^T - 2\mathbf{B}_{\text{AR2}}\Sigma_{2\tau})\mathbf{P}_\theta^T\}, \quad (31)$$

with $\Sigma_{1\tau} = \Sigma(\rho = v\tau)$ and $\Sigma_{2\tau} = \Sigma(\rho = 2v\tau)$.

Likewise for the AR3:

$$\begin{aligned} \sigma_{\text{lag}}^2(\tau, p = 3) = & \text{trace}\{\mathbf{P}_\theta(\Sigma_\varphi + \mathbf{A}_{\text{AR3}}\Sigma_\varphi\mathbf{A}_{\text{AR3}} + \mathbf{B}_{\text{AR3}}\Sigma_\varphi\mathbf{B}_{\text{AR3}} \\ & + \mathbf{C}\Sigma_\varphi\mathbf{C}_{\text{AR3}} - 2\mathbf{A}_{\text{AR3}}\Sigma_{1\tau} - 2\mathbf{B}_{\text{AR3}}\Sigma_{2\tau} - 2\mathbf{C}_{\text{AR3}}\Sigma_{3\tau} \\ & + \mathbf{A}_{\text{AR3}}\Sigma_{1\tau}\mathbf{B}_{\text{AR3}}^T + \mathbf{B}_{\text{AR3}}\Sigma_{1\tau}\mathbf{C}_{\text{AR3}}^T \\ & + \mathbf{A}_{\text{AR3}}\Sigma_{2\tau}\mathbf{C}_{\text{AR3}}^T)\mathbf{P}_\theta^T\}, \end{aligned} \quad (32)$$

with $\Sigma_{3\tau} = \Sigma(\rho = 3v\tau)$. The matrices $\Sigma_{p\tau}$ are computed for a fully developed turbulence following Eq. (13) and thus differ from those associated with any specific model.

In Fig. 5, the theoretical lag errors (for the cases **no** prediction and **with** prediction) are plotted [31]. Several comments follow:

- The one-step SA predictor provides the best performance (as expected) for large lags above ~ 5 ms, which stems from its optimality [it minimizes the prediction error variance in Eq. (16)] and cross-mode prediction. However, the predictor is only a truncated version; in practice, a finite number of modes is to be used. When more modes are added in, the one-step SA predictor beats the AR2-3 models for lower lags.

- For lags below ~ 5 ms AR models of orders 2 and 3 slightly outperform the one-step SA predictor since the short term decorrelation of the AR models is quite similar to that of the turbulence.

- The AR2 is a sufficient model-order above which the performance gains are little and not worth the increased complexity for both parameter identification or real-time processing; an AR2 model presents a temporal PSD $\propto \nu^{-4}$, which is a good approximation to the actual phase whose spectrum is $\propto \nu^{-11/3}$, with $\nu = |v|$ the temporal frequency vector modulus.

- Although the AR1 has been successfully used in control-oriented models for Kalman filtering [25,29,32], it performs quite poorly to predict phase and is patently incapable to provide any improvement over the **no** prediction case. Simulations show that when the off-diagonal values of the one-step SA predictor are nulled out, the predictive capability degrades to that of a diagonal AR1, suggesting that it is this feature that plays the most important role in the predictive process.

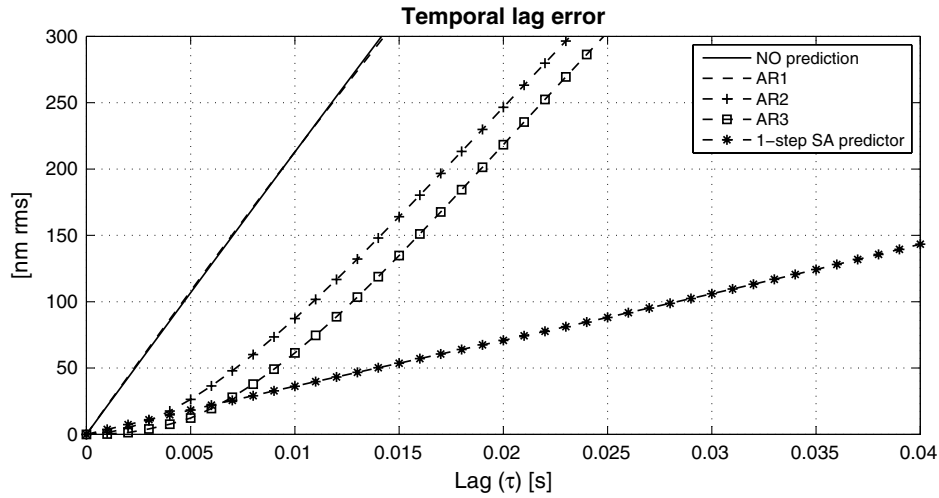


Fig. 5. Comparison of temporal lag errors on an equivalent single layer atmosphere (see Table 1 for further parameters).

Despite the results above, the offline computations of AR models in general are quite complex in fitting the autocorrelation functions (Section 3.C.1) and are much higher than the SA offline load, which does not require a fitting algorithm. Only addressing the synchronous case, the reconstructor dimensions are given by $r = N_\beta N_a N_s$ for the cases of no prediction, AR1 prediction, and SA prediction. With $p \geq 2$ one has $r = N_\beta N_z (N_a + p N_s)$. Thus the real-time computational load is not increased over a static reconstructor for AR1 and one-step SA prediction models. For the Raven case, whose parameters are given in Table 1, this corresponds to a 30% increase for the AR2 model and an 85% increase for the AR3 model.

4. INCREASING THE LIMITING MAGNITUDE FOR THE RAVEN PROJECT

Raven is a MOAO science and technology demonstrator, which will be the first MOAO instrument on an 8 m class telescope feeding an AO-optimized science instrument, the Subaru InfraRed Camera and Spectrograph (IRCS). The instrument will be equipped with three natural guide star (NGS), wavefront sensors (WFS), and one laser guide star (LGS) WFS to generate a tomographic reconstruction of the atmosphere in a 3.5 arcmin field of regard (FoR) for up to two science-object directions (see Fig. 6).

The specifics of the system are given below:

- **OL-WFSs:** Three deployable pick-off mirrors can patrol the telescope focal plane and pick off three NGS within a 3.5 ft FoR. Each pick-off mirror feeds one OL WFS consisting of a 10×10 SH-WFS ($d = 0.8$ m) with 12×12 pixels per subaperture and a 4.8 in. FoV per subaperture. The system's baseline detector is the E2V EMCCD camera. These devices can be set to run at very high gain. In this mode, the read noise can be made almost arbitrarily small; however, the background source plus background photon noise is increased.

- **An additional OL-WFS** with identical specifications is fixed on-axis in order to utilize the Subaru on-axis LGS, which will improve sky coverage and/or AO correction.

- **DMs:** There are two ALPAO DMs, one in each science channel, with 11×11 actuators and a 25 mm clear aperture.

These will contribute a constant fitting error term of 145 nm rms at $r_0 = 15.6$ cm.

Table 1 provides further simulation and system parameters.

A. Equivalence of Static Reconstructors

The equivalence of the two formulations for the static reconstructors presented in Section 2 is now established numerically. It boils down to checking the similarity between the

Table 1. Raven Baseline Configuration Parameters

Telescope	
D	8 m
Atmosphere	
r_0	15.6 cm
L_0	30 m
Zenith angle	0 deg
Fractional r_0	[0.596; 0.224; 0.180]
Altitudes	[0; 5; 10] km
Wind speeds	[7.5; 12.5; 15] m/s
Wind direction	[0; 0; 0] deg
Sampling	8/150 m
Wavefront Sensor	
RON	$0.2 e^-$
N_{NGS}	3
NGS radii	30 arcsec
Order	10×10
θ_{pix}	0.4 arcsec
N_{pix}	15
f_{sample}	30–200 Hz
λ_{WFS}	0.7 μm
Centroiding Algorithm Thresholded Center of Gravity	
DM	
Order	11×11
Stroke	Infinite
Influence	Cubic
AO loop	
Pure delay	$\Delta = 3$ ms
Controlled modes	$N_z = 55$
Reconstructor	$N_p = 406$ modes
Predictor	90 modes
Evaluation	
λ_{evl}	1.65 μm

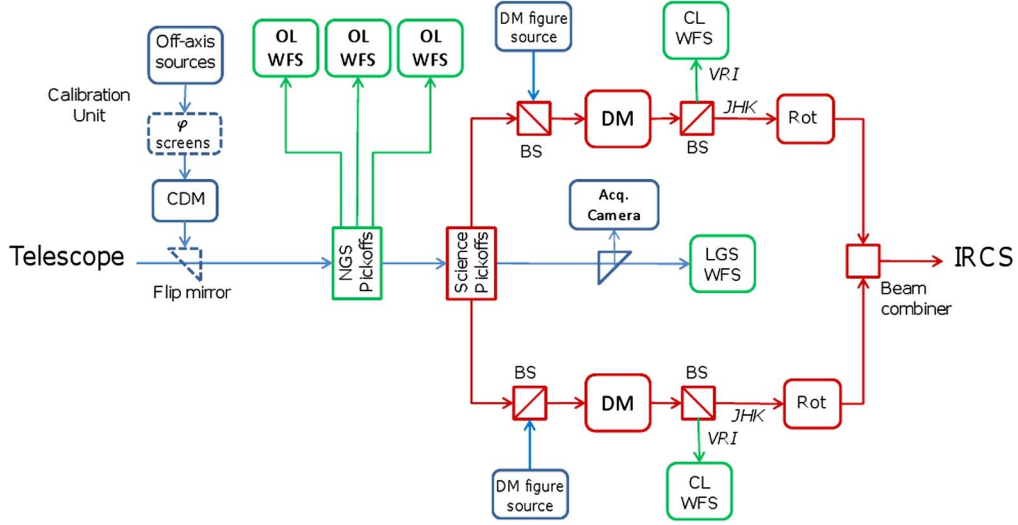


Fig. 6. Functional optical block diagram of RAVEN. Dashed blocks are deployable. Raven consists of eight main subsystems: the deployable calibration unit, the OL NGS WFSs, the science pick-offs, the science relays, the closed-loop NGS truth/figure WFSs, the beam combiner, the LGS WFS, and the acquisition camera.

angular covariance matrix between Zernike polynomials appearing in Eqs. (8b) and (11).

Their similarity increases with the increasing number of modes considered in each layer. Let ϕ_p be expanded onto a Zernike set of with a fixed number N_z of modes. The layered turbulence is expanded on a greater number of N_p modes.

It is instructive to assess the relative error

$$\varepsilon_i(N_p) = \frac{\text{trace}((\phi_\alpha \phi_\alpha^T) - \mathbf{P}_\alpha(N_p) \langle \phi \phi^T \rangle \mathbf{P}_\alpha(N_p)^T)}{\text{trace}(\langle \phi_\alpha \phi_\alpha^T \rangle)} \quad (33)$$

as a function of N_p to obtain a reasonable approximation to the SA covariance matrices computed using an analytical formulation. The relative error in Eq. (33) falls to the few-percent level when N_p is the result of using three times more radial orders than in N_z and below the 1% level for a factor of for, when $N_z = 55$ (nine radial orders [8]).

With this in mind, static reconstruction of a single-shot WF from slopes has been conducted on the Raven optical bench (Fig. 6, Table 1) [1] covering the full 2 arcmin FoV. The contour plots obtained are depicted in Fig. 7, demonstrating the full equivalence of performance obtained with the SA and the explicit reconstructors using Raven bench data. Differences are in the subpercent level.

B. Limiting Magnitude

The science gain achievable by Raven, in comparison to classical AO systems such as Subaru's AO188, will be modest because Raven will only have two science channels and provide a relatively low-order correction. Nevertheless, the 8 m aperture of the Subaru telescope enables science that is not achievable on smaller telescopes, and Raven will be capable of delivering high ensquared energy (EE) into the IRCS slit. It has been projected that, with a static tomographic reconstructor, Raven's limiting magnitude will approach 14.5 (for 30% EE) using a reduced frame rate of 180 Hz. As stated in [1], sky coverage for Raven will be low. For example, consider a point with galactic coordinates $(b; l) = (30; 0)$. Using the Besançon model of the galaxy [33],

one finds that there are 750 stars per square degree with $R < 14.5$ (1040 stars per square degree with $R < 15$). The probability that there are three stars with $R < 14.5$ in a 2 arcmin diameter FOR is just 3%. This does not even account for asteroids that are unsuitable for Raven; in some cases, the science targets will not be inscribed within the potential NGS asterism, and therefore the tomographic error will be too great. Star densities are increasing as a power law at these magnitudes, so going 1 magnitude deeper can increase the density of available stars by a factor of 1.8.

Results in the previous sections indicate that the potential gains in performance can be traded for an increase in limiting magnitude. In the presence of temporal prediction, the lag error σ_{lag}^2 decreases, and one can thus tolerate more measurement noise from fainter sources, resulting in increased noise propagation σ_{np}^2 (averaged over the field) for the same aggregate wavefront error, defined as

$$\sigma_{\text{total}}^2 = \underbrace{\sigma_{\text{lag}}^2 + \sigma_{\text{np}}^2}_{\sigma^2} + \text{other terms}, \quad (34)$$

where other terms relates to errors that are independent from the system lag (to a first degree approximation). The latter is computed from Eq. (28) whereas the noise propagation error is

$$\sigma_{\text{np}}^2 = \text{trace}\{\mathbf{E} \Sigma_\eta \mathbf{E}^T\}, \quad (35)$$

where the noise covariance matrix $\Sigma_\eta = \sigma_\eta^2 \mathbf{I}$ is assumed diagonal with σ_η^2 the measurement noise variance on each WFS sub-aperture. The latter is a decreasing function of the star brightness, i.e., photon-noise increases for dimmer guide stars. The propagated noise σ_{np}^2 relates to star magnitude by standard centroiding error functions [34].

The AO system bandwidth is chosen to minimize the error $\sigma^2 = \sigma_{\text{lag}}^2 + \sigma_{\text{np}}^2$. Thus one has

$$\sigma^2 = \sigma_{\text{np}}^2(m_v, \tau) + \sigma_{\text{lag}}^2(\tau, p = 0) = \sigma_{\text{np}}^2(m'_v, \tau) + \sigma_{\text{lag}}^2(\tau, p > 0), \quad (36)$$

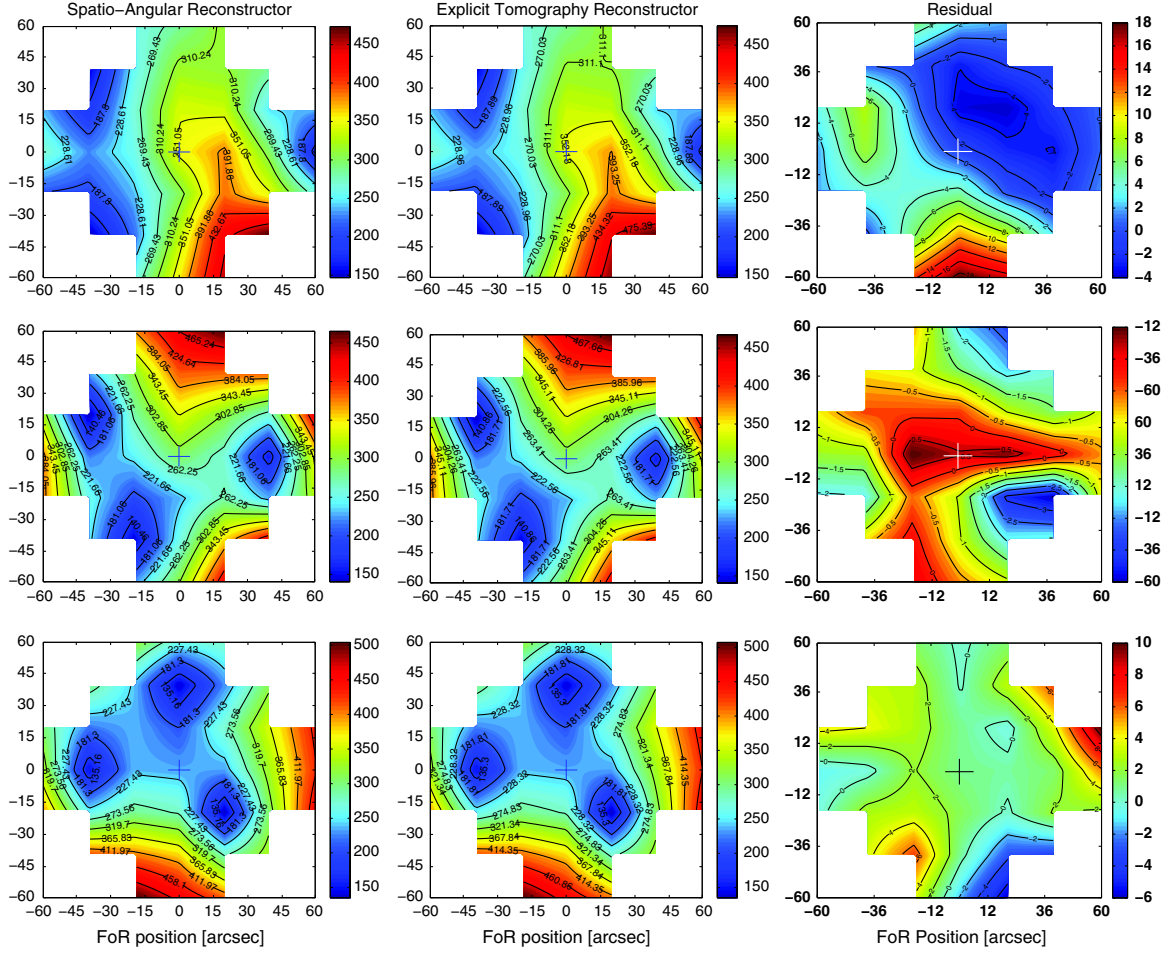


Fig. 7. Comparison of residual wavefront error (in nm RMS) across the FoR with the SA (left column) and the explicit tomographic reconstructor (central column), for NGS asterisms 2, 1.5, and 1 ft apart. Differences plot on the right column.

where $\sigma_{\text{lag}}^2(\tau, p = 0) \geq \sigma_{\text{lag}}^2(\tau, p > 0)$ and $\sigma_{\text{np}}^2(m_v, \tau) \leq \sigma_{\text{np}}^2(m'_v, \tau)$, with $m'_v - m_v$ the limiting magnitude increase.

This rather simplistic approximation offers insight into the potential magnitude increase factor. Figure 8 plots the result, $m'_v - m_v$, as a function of lag. The noise-propagation

coefficient used was computed to be $\sigma_{\text{np}}^2/\sigma_{\eta}^2 \approx 0.5$, following an extension to tomography of the analytical derivation in [35].

As expected, the minima in σ^2 (in blue, ordinate on the right) are achieved for longer integration times as the GS are fainter. These minima are indicated by vertical dotted

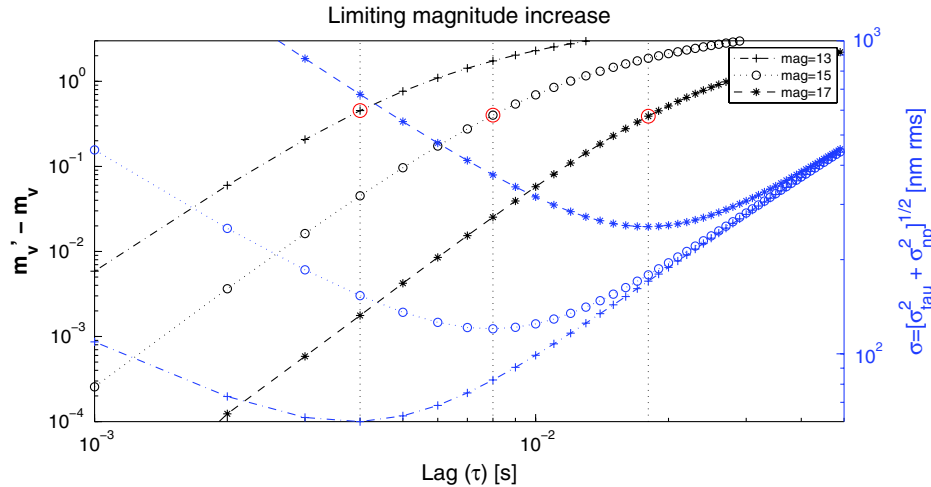


Fig. 8. Temporal lag error is traded by increased noise propagation through the wavefront reconstruction. Black markers: increase in limiting magnitude; blue markers: combined temporal plus noise propagation error; red circles indicate the increased limiting magnitudes for the minima of σ .

Table 2. Raven End-to-End Simulation Results^a

GS Mags	Static MV				Static MV + AR2				Static MV+ One-Step SA				Static MV + AR1			
	EE	Lag	Strehl	Lag	EE	Lag	Strehl	Lag	EE	Lag	Strehl	Lag	EE	Lag	Strehl	Lag
14.0	35.70	7	30.08	7	35.83	9	30.71	10	36.70	9	32.26	10	35.67	8	30.09	8
14.5	35.39	7	28.70	9	35.65	10	29.89	11	36.41	9	30.98	10	35.59	9	28.73	9
15.0	35.32	9	26.69	10	35.38	10	28.38	12	36.00	9	29.74	12	35.11	9	26.71	12
15.5	34.48	10	24.39	12	35.27	12	26.85	14	35.49	12	28.47	17	34.38	10	24.32	13
16.0	33.61	12	21.58	14	34.12	14	24.23	17	35.34	23	26.95	24	33.30	12	21.29	14
16.5	32.24	14	18.05	17	33.59	18	21.60	21	35.10	27	24.81	27	31.98	14	17.97	18
17.0	30.39	18	14.39	27	32.16	21	17.91	28	34.24	33	21.41	33	30.27	19	14.30	28

^aThe optimal performance (% ensquared energy, % Strehl ratio) for each GS magnitude is shown for three reconstructors: the static MV, SA prediction, and the AR2 prediction model. AR1 prediction is included for comparison purposes. Lags quoted in milliseconds.

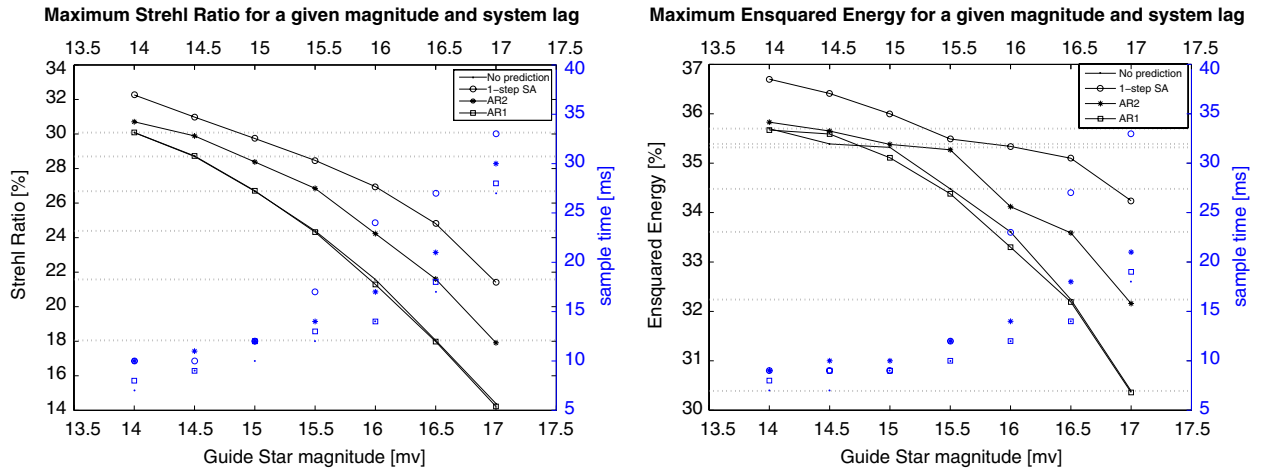


Fig. 9. Left: Strehl ratio. Right: ensquared energy. Peak performance achieved in simulation for GS magnitudes from 14 to 17 using each of four algorithms: static, SA prediction, AR2 prediction, and AR1 prediction (black) and the integration time in ms (ordinates on the right) at which the peak value was reached (blue).

lines, for which red circles indicate the increased limiting magnitudes when the sources vary from magnitudes 13 to 17. A consistent value around half a magnitude increase is obtained. However, only a full end-to-end Monte Carlo simulation can inform about the effective magnitude increase gain, considering the actual EE and Strehl-ratio (SR) figures of merit.

C. Sample Numerical Simulation

Results of Monte Carlo numerical simulations show the improvement in system performance using EE and SR as the benchmark figures of merit. Simulation parameters were selected to reflect the observing conditions of Raven with the telescope and atmosphere simulator. This includes an asterism of 3 NGSS within a 2 arcmin FOR; in this case, an asterism with a diameter of 0.5 arcmin was selected in order to reduce tomographic errors and highlight the temporal aspects of system performance. The full set of simulation parameters are listed in Table 1.

The results of the numerical simulations are shown in Fig. 9 and summarized in Table 2. Following Section 4.A, turbulence is expanded on 27 radial orders (i.e., 406 modes) from which nine (55 modes) are controlled. For computational reasons, the predictive models are computed for 90 out of 406 modes per layer with no significant effect on the overall performance.

The data shows the peak system performance in SR and EE, on the Subaru IRCS 140 mas slit, for an asterism of three guide stars of the same magnitude, and the corresponding sampling rate at which the peak occurs for each of three algorithms: static MV, static MV using an AR2 model, and static MV using the one-step SA predictive model. For interest and comparison purposes, the AR1 model also was tested. Lags corresponding to frame rates of 30–200 Hz were used. The simulation incorporates the fixed system lag of 3 ms allotted for camera read-out, data processing, and issuing of DM commands. In order to model the delayed application of the DM commands in the middle of an exposure, the system is attributed a fixed sample rate of 1 kHz. The resulting output phase is summed over the first 3 ms (three samples) before the new DM command is applied; subsequently, the output phase for the remaining total exposure time is added to these first frames to make a total exposure of the desired length. Each simulation run collected 2000 exposures before computing the SR and EE.

The results confirm that a reduction in frame rate, combined with SA temporal prediction (as noted in [11]) will allow the system to achieve a level of performance for a given GS magnitude, which is equal to the performance with a static reconstructor using GSs one magnitude brighter, and the performance with an AR2 predictive reconstructor using GSs half a magnitude brighter. This achievement was underestimated

through the somewhat simplistic model of Eq. (34) used in Fig. 8, for the noise is not spatially invariant, its propagation through the reconstruction isn't constant for varying magnitudes, and neglected terms do indeed depend upon the lag. The results also confirm the computations shown in Fig. 5 that estimate no reduction in temporal error with the use of an AR1 prediction model.

5. CONCLUSION AND OUTLOOK

This paper explores the features of two mathematically equivalent versions of algorithms to perform tomography in wide-field AO systems: the commonly used minimum-variance estimators with explicit 3D reconstruction of turbulence profile and the MOAO-specific SA reconstructors adopted in the Learn & Apply from [4] where the wavefront is directly estimated in the pupil plane.

A merge of both formulations is proposed to overcome (1) a challenging calibration since MOAO systems operate in OL and (2) intrinsic temporal lag errors by embedding a predictive model to work with fainter sources and thus increase the sky coverage. The former is tackled by partially computing the tomographic reconstructor from OL measurements. This mitigates issues related to under modeling and unknown systematics between the WFSs. The latter needs always to be model-based, with parameters identified by postprocessing the measurements' covariance matrix with the built-in SLODAR method. This renders the tomographic reconstruction a highly data-driven, self-sufficient approach to circumvent the complex calibration in OL operation.

Several predictive models are described along with a much faster and computationally sound identification procedure over previous methods (assuming Zernike modes). The predicted error structure functions are analytically derived. Diagonal AR models of order 2 are shown to perform equally well as the SA predictor model for small lags below ≈ 10 ms. For larger lags, the one-step SA predictor is preferred.

These models can now be plugged into the linear-quadratic-Gaussian controller, which is expected to push further the limiting magnitude. At this stage, the wind profiles are assumed known, and only pure frozen flow is simulated. In reality, the profiles are estimated and frozen flow may not be always present, which calls for a robustness assessment to be done in a forthcoming paper.

For instance, on the Raven project, results in this paper suggest the same level of performance is achieved by combining temporal prediction with guide stars 1 magnitude fainter.

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